1	2	3	4	5	6	Sum
/20	/20	/20	/20	/20	/20	/100

Name______________________________ Jan 15, 2016 Friday

SID________________________________ School/Department______________________

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu)=\int_0^{\infty} e^{-t}t^{\nu-1}dt,\ \Gamma(\nu+1)=\nu\Gamma(\nu),\ \Gamma(1)=1,\ \Gamma(1/2)=\sqrt{\pi}.$$

c	h	e	m_e	u	N_A	k_B
3.0×10^8 m/s	6.63×10^{-34} J·s 1.24×10^3eV ·nm/c	1.6×10^{-19} C	9.11×10^{-31} kg 0.511MeV/c^2	931.494MeV/c^2	6.02×10^{23} mol^{-1}	8.617×10^{-5}eV/K

The quark combinations of some particles: p—uud n—udd Λ^0 – uds Σ^+ – uus

Quantum numbers of three quarks:

Quark symbols	Charge numbers	Spin	Baryon numbers	Strangeness
u	2/3	1/2	1/3	0
d	−1/3	1/2	1/3	0
s	−1/3	1/2	1/3	−1

Select five out of the following six problems.

1.(20 pts.) When neutrons are in thermal equilibrium within a reactor, they satisfy the equipartition theorem.

(a) If the temperature of the reactor is 300 K, what is the average kinetic energy $\bar{\varepsilon}$ of such a thermal neutron?

(b) The mass of neutron is known to be 1.0087 u, What is the corresponding de Broglie wavelength λ ?

2. (20 pts.) After long effort, in 1902, Marie and Pierre Curie succeeded in separating from uranium ore the first substantial quantity of radium, one decigram of pure $RaCl_2$(radium chloride). Actually the radium they obtained was the radioactive isotope ^{226}Ra, which has a half-life of 1600 a.

(a) How many nuclei of ^{226}Ra had they isolated?

(b) What is the average life time τ of the isotope ^{226}Ra.

(c) What is the decay rate R of their sample at a late time t?

(The atomic mass of Cl(chlorine) is 35.5u .)

3.(20 pts.) An electron of a stationary hydrogen atom passes from the fifth level to the ground state, with a photon emitted in the process.

(a) Calculate the energy E_1 of the ground state and the energy E_5 of the fifth level, according to the Bohr theory.

(b) Considering the recoil of the hydrogen atom, what is the energy E_γ of the emitted photon, to the order of $(E_5 - E_1)^2 / mc^2$ where $m = 938.782\,\mathrm{MeV}/c^2$ is the rest mass of the hydrogen atom?

(c) To the same order of magnitude, what is the recoil energy E_{recoil} of the hydrogen atom?

4.(20 pts.) Black body radiation can be treated as an ideal gas of photons. As such, its density of states $D(\nu)$ in the frequency ν space is $D(\nu)=V8\pi\nu^2c^{-3}$ and the number of quantum states within the interval of frequency ν to $\nu+d\nu$ is $V8\pi\nu^2c^{-3}d\nu$, where V is the volume of the photon gas.

(a) Show that the internal energy of the photon gas is $U=\alpha T^4V$ where T is temperature of the system and $\alpha=k_B^4\Gamma(4)\varsigma(4)/\pi^2c^3\hbar^3$.

(b) Find the heat capacity at constant volume C_V of the system.

(c) Find the Helmholtz free energy F and the Gibbs free energy G of the system.

(The fundamental equation of thermodynamics here can be written as $dU=TdS-pdV$.

$\int_0^{+\infty}\frac{x^{\nu-1}}{e^x-1}dx=\Gamma(\nu)\varsigma(\nu),\nu>0.$)

5. (20 pts.) A one-dimensional simple harmonic oscillator with mass m and natural angular frequency ω is found to stay in the quantum state of following normalized wave function

$$\psi=c\left(\frac{\alpha}{\sqrt{\pi}}\right)^{1/2}e^{-\frac{1}{2}\alpha^2x^2}+\left(\frac{\alpha}{\sqrt{\pi}}\right)^{1/2}\alpha x e^{-\frac{1}{2}\alpha^2x^2},$$

where $\alpha=(m\omega/\hbar)^{1/2}$, the c is a positive constant to be determined.

(a) Determine the constant c.

(b) Is the ψ an eigenfunction of the Hamiltonian of the oscillator?

(c) Find $\langle H\rangle$, the average energy of the oscillator.

(d) Find ΔE, the uncertainty or fluctuation of the energy of the oscillator.

6. (20 pts.) For each of the following reactions state whether or not it is forbidden. If it is forbidden, state a conservation law that is violated.

(a) $\Lambda^0 \to p + \pi^0$

(b) $\overline{p} + p \to \mu^+ + e^-$

(c) $n \to p + e^- + \nu_e$

(d) $n + \nu_e \to p + e^-$

(e) $\Sigma^+ \to \Lambda^0 + \pi^+$